

Representing Syntax: BNF Grammars

Syntax: form of a language

Semantics: meaning of a language

Two main ways of representing syntax:

- BNF Grammars
(Context-Free Grammars)
- Syntax Diagrams
(Like finite automata, but with some differences)

BNF Grammars

Backus Naur Form grammars

John Backus: FORTRAN, ALGOL, IBM Speedcode.
1977 Turing Award winner. (Philadelphia, US)

Peter Naur: ALGOL. 2005 Turing Award winner. (Denmark)

Components of a BNF grammar:

- Terminals
- Nonterminals
- Start symbol (will be one of the nonterminals)
- Productions (rules)

Example:

$$\begin{aligned}\langle \text{expr} \rangle &\rightarrow \langle \text{item} \rangle + \langle \text{item} \rangle \\ \langle \text{expr} \rangle &\rightarrow \langle \text{item} \rangle - \langle \text{item} \rangle \\ \langle \text{item} \rangle &\rightarrow X \\ \langle \text{item} \rangle &\rightarrow Y\end{aligned}$$

Terminals: $X, Y, +, -$

Nonterminals: $\langle \text{expr} \rangle, \langle \text{item} \rangle$

Start symbol: $\langle \text{expr} \rangle$

A sample derivation:

$$\begin{aligned}&\langle \text{expr} \rangle \\ \Rightarrow &\langle \text{item} \rangle + \langle \text{item} \rangle \\ \Rightarrow &X + \langle \text{item} \rangle \\ \Rightarrow &X + Y\end{aligned}$$

Another sample derivation:

$$\begin{aligned}&\langle \text{expr} \rangle \\ \Rightarrow &\langle \text{item} \rangle - \langle \text{item} \rangle \\ \Rightarrow &\langle \text{item} \rangle - Y \\ \Rightarrow &Y - Y\end{aligned}$$

Bigger Example:

$\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle + \langle \text{expr} \rangle$ $\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle - \langle \text{expr} \rangle$ $\langle \text{expr} \rangle \rightarrow \langle \text{item} \rangle$ $\langle \text{item} \rangle \rightarrow X$ $\langle \text{item} \rangle \rightarrow Y$

Derivation of **X**:

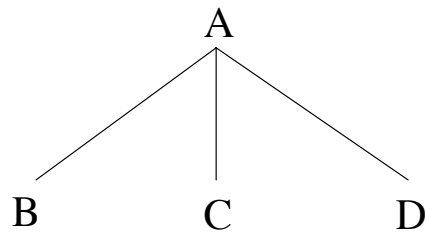
$$\begin{aligned} & \langle \text{expr} \rangle \\ \Rightarrow & \langle \text{item} \rangle \\ \Rightarrow & X \end{aligned}$$

Derivation of **X - Y - X**:

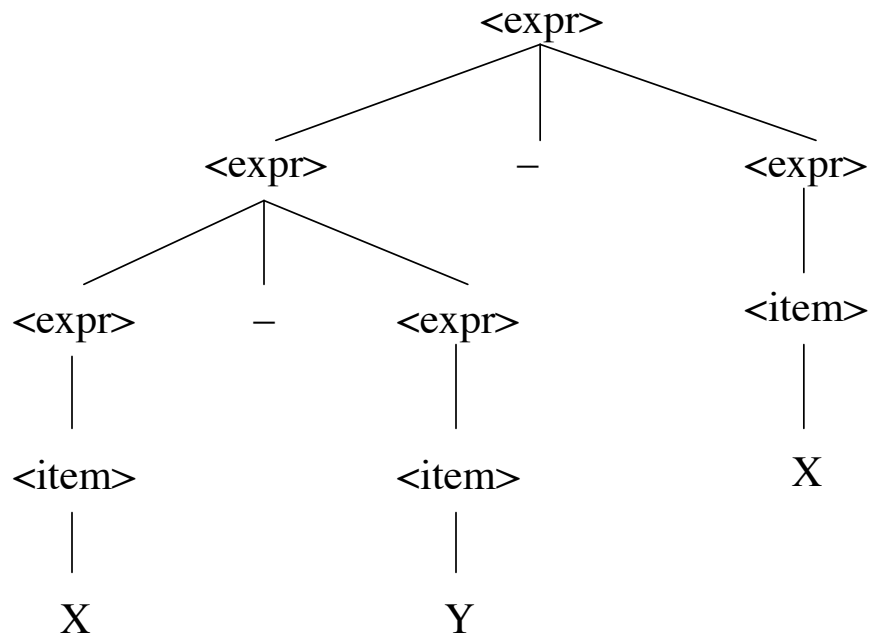
$$\begin{aligned} & \langle \text{expr} \rangle \\ \Rightarrow & \langle \mathbf{expr} \rangle - \langle \text{expr} \rangle \\ \Rightarrow & \langle \mathbf{expr} \rangle - \langle \mathbf{expr} \rangle - \langle \text{expr} \rangle \\ \Rightarrow & \langle \text{item} \rangle - \langle \text{expr} \rangle - \langle \text{expr} \rangle \\ \Rightarrow & \langle \text{item} \rangle - \langle \text{item} \rangle - \langle \text{expr} \rangle \\ \Rightarrow & \langle \text{item} \rangle - \langle \text{item} \rangle - \langle \text{item} \rangle \\ \Rightarrow & X - \langle \text{item} \rangle - \langle \text{item} \rangle \\ \Rightarrow & X - Y - \langle \text{item} \rangle \\ \Rightarrow & X - Y - X \end{aligned}$$

Parse Trees

Idea: $A \rightarrow B C D$ represented as

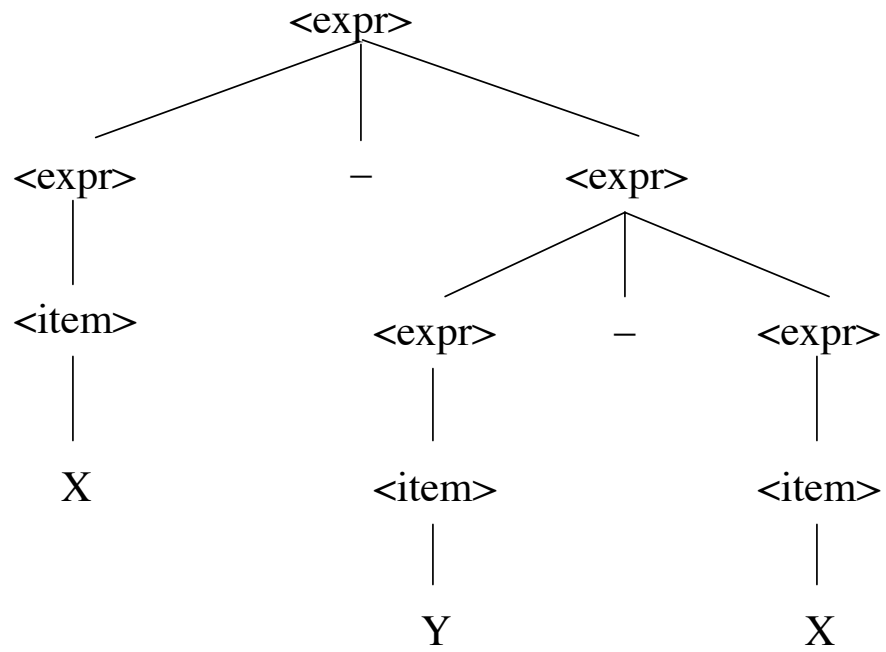


Parse tree for **X - Y - X**



There is a different parse tree for **X - Y - X**. What is it?

Different parse tree for **X - Y - X**



Grammar:

$\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle + \langle \text{expr} \rangle$ $\langle \text{expr} \rangle \rightarrow \langle \text{expr} \rangle - \langle \text{expr} \rangle$ $\langle \text{expr} \rangle \rightarrow \langle \text{item} \rangle$ $\langle \text{item} \rangle \rightarrow X$ $\langle \text{item} \rangle \rightarrow Y$

Question: The string **X - Y - X** can be derived from this grammar with two different parse trees. What is the term for grammars with this property?

Challenge: Fix the grammar.

A grammar for a subset of Scheme lists:

```
<list> → ( )  
<list> → ( <sequence> )  
<sequence> → <element> <sequence>  
<sequence> → <element>  
<element> → X  
<element> → Y
```

Question: Here are some strings that I want to be able to derive:

```
( X )  
( X Y X )  
( )  
(( X ))  
( X ( Y X ) )
```

Which of these strings can be derived from my grammar?

Challenge: Fix my grammar so that it can derive all the other strings too.

Extended BNF

Extended BNF notation adds some extra symbols, for convenience:

- | Alternatives
- [] Optional element: 0 or 1 occurrences of the item in brackets
- { } Repeated element: 0 or more occurrences of the item in braces

Examples:

$\langle \text{item} \rangle \rightarrow X \mid Y$

$\langle \text{sequence} \rangle \rightarrow \langle \text{element} \rangle \mid \langle \text{element} \rangle \langle \text{sequence} \rangle$

$\langle \text{sequence} \rangle \rightarrow \langle \text{element} \rangle [\langle \text{sequence} \rangle]$

$\langle \text{sequence} \rangle \rightarrow \langle \text{element} \rangle \{ \langle \text{element} \rangle \}$

$\langle \text{list} \rangle \rightarrow (\{ \langle \text{element} \rangle \})$